

CHLOE HALLAERT

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TECHNICAL SKILLS

Languages Python, C++, C, JavaScript, HTML
Technologies Docker, PyTorch, Azure, AWS, TensorFlow, Pandas, NumPy, SQL, Unix, Linux, Git, ROS2, MILVUS

EDUCATION

Bachelor of Software Engineering, Honors — University of Waterloo

University of Waterloo

- Coursework: Object-Oriented Programming, Data Structures & Algorithms, Linear Algebra, Statistics, Probability 2024-Present
- Awarded International Student Scholarship (one of 20 recipients) and President's Scholarship

EXPERIENCE

AI & Robotics Software Engineer | Python, PyTorch, NumPy, ROS, Isaac Sim, Docker, MLflow

Jan-May 2026

Mara Inc

San Francisco, California, United States

- Improved transformer-based drone tracking model (**MixFormer**) **performance by 350%** by redesigning the scoring module with custom loss, label smoothing, fine-tuned weights, and template-update guards, then shipped it into the production pipeline
- Generated **hundreds of labeled UAV training scenarios** by developing custom **Isaac Sim** environments with URDF-modeled objects and RL-driven robot behaviors, improving model robustness and accelerating synthetic data generation
- Reduced drone dataset preparation **time by 97%** by building a **dataset management CLI** with automated GPS/NavLite time-range detection, clip generation controls, spherical-coordinate labeling, and self-updating metadata for training pipelines

AI Project Developer | Azure AI, Machine Learning, Cloud Computing

May-Aug 2025

Accenture

Toronto, Ontario, Canada

- Led a team of 5 to design and present "TrustScan," a browser extension that detects fraudulent web activity, integrating a supervised logistic regression model and Azure AI Language to deliver chatbot-based scam warnings

AI & Software Engineer | Python, AWS, Docker, Milvus, Selenium, BS4, SQLite, SQL, Transformers

Sept 2024-July 2025

Wilson & Wilbur Consulting Group

Toronto, Ontario, Canada

- Integrated the **LLaMA 1B** model into a **Dockerized chat service**, exposing API endpoints for conversation, summarization, and reasoning; designed prompts, evaluated outputs with **Transformers/evaluate**, and implemented unit tests
- Developed a Retrieval-Augmented Generation (**RAG**) **pipeline** and deployed a vector data service using **Milvus** and **Docker**, implementing CRUD operations on vector databases to improve efficiency and reduce upload time by **70%**
- Built and deployed a **web scraper** using **Selenium** and **Beautiful Soup**, leveraging multithreading and multiprocessing to scale automated data ingestion, implemented unit tests, deployed to **AWS EC2**, and optimized data extraction with **SQL/SQLite**

AI & Robotics Software Engineer | Python, PyTorch, TensorFlow, ROS2, YOLOv5, pandas

July 2022

University of Maryland

College Park, Maryland, USA

- Developed a object detection and semantic navigation system using **Python**, **PyTorch**, **TensorFlow**, and **ROS2** to enable real-time perception, tracking, and probabilistic semantic mapping for ground robot navigation [🔗\[demo\]](#)
- Built a **custom neural network** and integrated **YOLOv5** and a custom binary classification model for real-time detection and adaptive navigation; fused sensor and visual input data using **pandas** for streamline data manipulation

PROJECTS

Synthesizer and Musical Keyboard | C, Raspberry Pi, FluidSynth API

- Collaboratively developed a one-octave musical keyboard using a **Raspberry Pi Zero** and **FluidSynth API**, implementing GPIO interrupts in **C** to enable real-time audio playback, improve software reliability, and enhance responsiveness

LEADERSHIP

Assistant Director & Math and English Tutor | Teaching, Student Engagement

Aug 2022 - April 2024

Kumon Math and Reading Center of Bethesda - Kenwood

Maryland, USA

- Led one-on-one tutoring for 80+ math and English students, tailoring learning plans based on progress and needs
- Communicated regularly with clients, gave progress updates, addressed concerns, and offered advice for improvement

ADDITIONAL

First Honors - Received award for academic excellence every semester

Brown University - Mathematical Modeling of Finance: An Introduction to Quantitative Analysis